Large Language Models for Time Series Analysis: Methodologies, Applications, and Emerging Challenges

Wangzhi Yu, Dawei Cheng and Lizhao Zhu

***Abstract*—Large language models (LLMs), pre-trained on massive corpora have driven remarkable breakthroughs in natural language processing and computer vision. Building upon these successes, recent studies have examined the potential of pre-trained LLMs for time series analysis (TSA), focusing on downstream tasks such as time series forecasting, classification, anomaly detection and imputation across domains including finance, traffic and energy. In this survey, we systematically categorize the roles of LLMs for TSA into three principal types and further abstract their unified workflows: (1) fine-tune-based inference engines, where LLMs serve as core models for TSA tasks, with workflows involving tokenization, prompt engineering, semantic alignment, and fine-tuning; (2) TSA-methods-based enhancers, where LLMs serve as auxiliary modules within TSA methods; (3) hybrid collaborators, representing emerging approaches that integrate LLMs with lightweight models and multi-agent systems to enable advanced collaborative solutions. Then, we systematically review how LLMs empower downstream tasks across diverse domains. Specifically, we organize these applications according to the underlying mechanisms through which LLMs augment domain-specific tasks. This approach provides clear insights into the multifaceted roles of LLMs in advancing TSA across industries. Finally, we analyze the principal challenges faced by LLMs for TSA, including limitations such as cross-domain generalization, privacy preservation, and multimodal data alignment. By establishing a rigorous foundation, we aim to guide future research efforts and foster methodological innovation within the field.**

***Index Terms*—Time series, large language models, deep learning, time series analysis.**

I. Introduction

TIME series data refers to sequences of data points that are collected or recorded discretely at specific temporal indices [1], [2]. In recent years, there has been a substantial proliferation of time series data across a wide spectrum of domains, including finance, energy, and transportation. This expansion has underscored the imperative for time series analysis (TSA), which facilitates the systematic extraction and interpretation of temporal patterns and semantics in time series data [3], thereby providing a foundation for a variety of downstream tasks, such as forecasting [4], classification [5], anomaly detection [6], and imputation [7]. Consequently, the development of efficient and accurate TSA techniques is indispensable for enabling data-driven decision-making and advancing intelligent services.

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Researchers have explored a wide range of lightweight models for TSA, including statistical models [8], machine learning models [9], [10], deep learning models such as multi-layer perceptrons (MLPs) [11], [12], convolutional neural networks (CNNs) [13], [14], recurrent neural networks (RNNs) [15], and transformers [16], [17]. They have been widely used to extract essential temporal characteristics from time series data, including the periodic or seasonal patterns, and dynamic dependencies. Despite these effectiveness, these approaches still suffer from limited adaptability, as they are typically designed for specific downstream tasks and domains. The recent development of large-scale models has led to remarkable breakthroughs in artificial intelligence, particularly in natural language processing (NLP) and computer vision (CV). The representative model such as generative pre-trained transformer (GPT) [18], bidirectional encoder representations from transformers (BERT) [19], and contrastive language–image pre-training (CLIP) [20], powered by large-scale pre-training and fine-tuning paradigms, have achieved outstanding performance in text understanding and generation, image recognition, and more. Building upon the recent advancements in large models, some studies have focused on constructing large-scale time series datasets and developing time series foundation models (FMs) for TSA [21], [22].

However, FMs for TSA still encounter notable limitations in terms of generalization [1]. Specifically, time series data often exhibit pronounced heterogeneity in data distribution and format across different domains. For instance, financial time series are typically marked by high volatility and non-stationarity, while traffic-related time series are subject to periodic peaks, sudden disruptions, and intricate spatial dependencies. Such domain-specific characteristics severely limit the transferability of data representations and models across domains, resulting in suboptimal cross-domain generalization. Furthermore, FMs that are tailored to specific tasks such as forecasting, anomaly detection, or classification often lack the flexibility to be directly applied to other types of time series tasks, which leads to poor cross-task generalization. Recent studies have investigated the pre-training and fine-tuning paradigms of large languages models (LLMs) for TSA [23], [24], in which LLMs pre-trained on large-scale NLP corpora are further fine-tuned for various domains and tasks. This paradigm not only effectively addresses the significant adaptability limitation of lightweight models for specific downstream tasks within particular domains, but also overcomes the generalization challenges faced by FMs.

As illustrated in the framework of this survey presented in Figure 1, we first review the historical development of TSA models in Section II, tracing their evolution from lightweight models to FMs. Besides, in contrast to previous surveys [1], [2], we provide a systematic categorization of recent research based on the primary roles that LLMs play in TSA and their unified workflows, as detailed in Section III, encompassing fine-tune-based inference engines, TSA method-based enhancers, and hybrid collaborators. Furthermore, while previous surveys have primarily focused on the application of LLMs to specific tasks or domains [25], our work adopts a broader perspective by exploring how LLMs empower a wide range of downstream tasks and domains. Specifically, we systematically review the applications of LLMs across sectors such as finance, transportation, and energy, and categorize these applications according to the distinct mechanisms through which LLMs enhance domain-specific tasks, including multi-modal data fusion, decision support, and interpretability enhancement, as detailed in Section IV. This comprehensive approach provides clear insights into the diverse and multifaceted roles of LLMs in advancing TSA methodologies. Finally, we identify the remaining challenges in the field and discuss promising directions for future research in Section V. To further support research and practical implementation, we also curate and release a GitHub repository[1] that exclusively presents the categorization of papers as designed in this survey, providing a structured reference for the community. In summary, the main contributions of this survey are as follows:

- ***Taxonomy: Roles-based Unified Workflows.*** We systematically categorize the roles of LLMs for TSA and abstract unified workflows for each role, providing a standardized framework that elucidates their essential mechanisms.
- ***Empowerment: Applications in Downstream Domains and Tasks.*** We provide a comprehensive review of representative applications spanning multiple domains. These applications are systematically categorized according to the specific mechanisms by which LLMs empower and enhance domain-specific tasks. This approach offers new perspectives that can drive further advancements in downstream applications.
- ***Challenges: Limitations and Future Directions.*** We critically examine the key limitations and challenges in deploying LLMs for TSA, and further propose prospective research directions to address these challenges and advance the field.

II. Preliminary and Background

This section provides a concise overview of the main tasks in TSA with formal definitions and summarizes recent advances in lightweight models and FMs for TSA.

A. Preliminary

1) Time Series Data: Time series data refer to a collection of observations sampled at regular intervals, typically represented as $\mathbf{X} \in \mathbb{R}^{N \times T \times F}$. The i-th time series ($1 \leq i \leq N$) consists of F features (channels) observed over T historical time steps. Time series data can be categorized as follows:

[1]https://github.com/TongjiFinLab/awesome-time-series-forecasting.

Definition 1 (Univariate Time Series, UTS): When $N = 1$ and $F = 1$, the data represent a UTS, which corresponds to a single sequence of scalar observations over time.

Definition 2 (Multivariate Time Series, MTS): When $N = 1$ and $F > 1$, the data form a MTS, where each time step contains a vector of multiple features.

Definition 3 (Spatiotemporal Data, STD): When $N > 1$ and $F = 1$, the data are referred to as STD, comprising multiple univariate time series from different entities or locations.

Definition 4 (Multivariate Spatiotemporal Data, MSTD): When $N > 1$ and $F > 1$, the data are termed MSTD, representing multiple distinct multivariate time series collected from different entities or locations.

2) Time Series Analysis Tasks: We summarize four representative downstream tasks in TSA: forecasting, classification, anomaly detection, and imputation.

Definition 5 (Time Series Forecasting): Given historical data $\mathbf{X}_{1:T} \in \mathbb{R}^{N \times T \times F}$, the goal is to predict the next τ steps, i.e., to estimate $\hat{\mathbf{X}}_{T+1:T+\tau}$ using a forecasting function f_{fore}:

$$\hat{\mathbf{X}}_{T+1:T+\tau} = f_{\text{fore}}(\mathbf{X}_{1:T}) \tag{1}$$

Definition 6 (Time Series Classification): Given $\mathbf{X}_{1:T} \in \mathbb{R}^{N \times T \times F}$, the objective is to assign class labels $\hat{\mathbf{Y}} \in \mathcal{C}^N$ via a classification function f_{cls}:

$$\hat{\mathbf{Y}} = f_{\text{cls}}(\mathbf{X}_{1:T}) \tag{2}$$

Definition 7 (Time Series Anomaly Detection): Given $\mathbf{X}_{1:T} \in \mathbb{R}^{N \times T \times F}$, the task is to detect anomalies by estimating the anomaly indicator matrix $\hat{\mathbf{A}}_{1:T} \in \{0,1\}^{N \times T}$ (where 1 denotes anomaly and 0 denotes normality) using f_{anom}:

$$\hat{\mathbf{A}}_{1:T} = f_{\text{anom}}(\mathbf{X}_{1:T}) \tag{3}$$

Definition 8 (Time Series Imputation): Given incomplete data $\mathbf{X}_{1:T}$ and a binary mask $\mathbf{M}_{1:T} \in \{0,1\}^{N \times T \times F}$, the aim is to recover the complete time series data $\hat{\mathbf{X}}_{1:T}$ using an imputation function f_{imp}:

$$\hat{\mathbf{X}}_{1:T} = f_{\text{imp}}(\mathbf{X}_{1:T}, \mathbf{M}_{1:T}) \tag{4}$$

B. Background: Lightweight Models and Foundation Models

Recent advances in TSA have primarily focused on two major categories of models: lightweight models and FMs. In the following sections, we first introduce the representative architectures and key characteristics of each model type, and then systematically examine their main limitations.

1) Lightweight Models: Lightweight models have gained widespread adoption in TSA due to their low computational requirements and task-specific effectiveness. These models are typically tailored for specific forecasting scenarios, making them efficient for deployment in resource-constrained or real-time settings. Architecturally, lightweight models span several families, including MLPs, CNNs, RNNs and Transformer-based designs. Variants of MLPs [11], [12] and CNNs [13], [14] have shown strong capabilities in feature extraction and representation learning across various TSA tasks. With the advancement of research, more expressive models based on RNNs and Transformers have become prominent. RNN-based variants, such as vanilla RNNs [15], gated recurrent

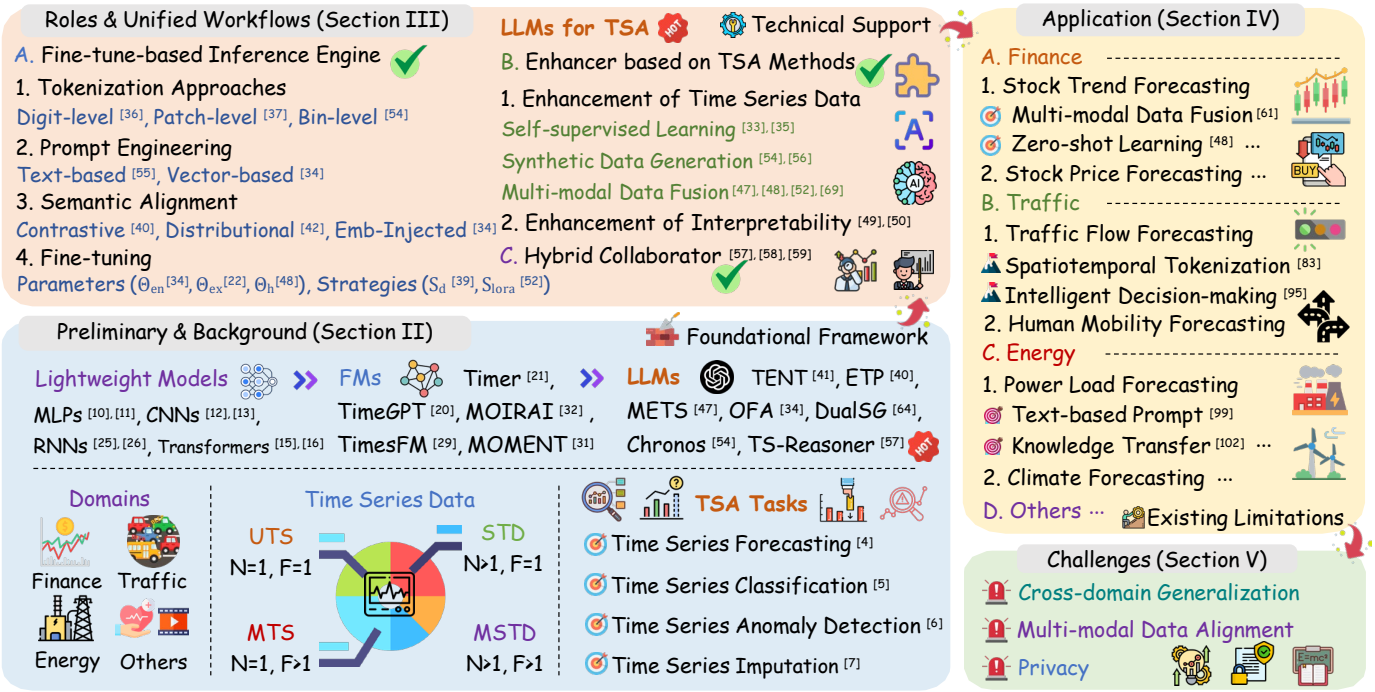

Fig. 1. The framework of this survey. Section II presents an overview of time series data and TSA preliminaries, and traces the evolution of TSA models from lightweight models to FMs and LLMs, thereby establishing the conceptual groundwork for the subsequent discussion. Section III offers a systematic classification of the principal roles of LLMs in TSA, encompassing fine-tune based inference engines, method-based enhancers, and hybrid collaborators, along with their unified workflows. Building on this technical foundation, Section IV explores how LLMs empower downstream domains and tasks, with representative applications in finance, transportation, and energy. Finally, Section V identifies key limitations and outlines open research directions, providing a structured basis for future methodological advancements in this field.

units (GRUs) [26], and long short-term memory (LSTM) networks [27], leverage recurrent connections and memory mechanisms to capture temporal dependencies in sequential data, while GRUs offer a more efficient structure. Transformer-based variants further improve scalability and parallelization, thereby making them well-suited for efficiently handling long sequences and large-scale datasets.

Inspired by the remarkable success of the Transformer architecture in NLP, an increasing number of studies have explored its adaptation to time series forecasting. The vanilla Transformer [28] employs self-attention mechanisms to capture global dependencies without relying on recurrence, thereby enabling superior parallelization and representation learning. However, its computational complexity grows quadratically with sequence length, which significantly limits its scalability. To overcome this limitation, several variants have been proposed. Informer [29] introduces a ProbSparse attention mechanism that effectively reduces the computational cost of attention, making long-sequence forecasting feasible. iTransformer [17] reconsiders positional encoding and input formatting to better accommodate irregular time series. PatchTST [16] further innovates by treating time series as patches, in a manner similar to vision transformers, and demonstrates strong performance across diverse forecasting benchmarks. Collectively, these models represent a shift toward more flexible and scalable architectures, laying a solid foundation for the emergence of large models in TSA. Despite the significant progress achieved by lightweight models in time series forecasting, they still exhibit limited adaptability across various tasks, domains, and data distributions. Most of these models are tailored for specific forecasting scenarios and often require task-specific architectural adjustments or retraining when applied to new datasets. Their performance tends to degrade substantially under distribution shifts, sparse supervision, or in situations with limited training data. Furthermore, these models struggle to generalize across heterogeneous types of time series or to incorporate domain-specific semantics without manual feature engineering.

2) Foundation Models: Large models for TSA can be broadly categorized into two main types: FMs and LLMs. FMs are typically pre-trained on large-scale time series corpora and optimized for general-purpose tasks, whereas LLMs originate from the NLP domain and offer broader multi-modal and semantic reasoning capabilit. FMs are large-scale models pre-trained on diverse and massive time series corpora, enabling them to learn generalizable patterns across tasks and domains. In TSA, FMs have shown promising performance, especially in zero-shot and few-shot settings, by reducing reliance on task-specific designs. TimeGPT [21] is trained on over 100 billion data points from domains like finance and meteorology, supporting both forecasting and anomaly detection. Timer [22] adopts a GPT-style architecture to unify heterogeneous time series into a standardized sequence format. TimesFM [30] focuses on decode-only forecasting and supports variable-length outputs on unseen datasets. Lag-Llama [31] targets univariate probabilistic forecasting using lag-aware covariates and achieves strong general-purpose performance. MOMENT [32] introduces a family of open-source FMs trained on the

time series pile dataset, demonstrating strong performance across multiple TSA tasks with minimal task-specific fine-tuning. MOIRAI [33] is Salesforce's masked encoder-based universal time series forecasting transformer that tackles cross-frequency learning and variable accommodation challenges.

III. Taxonomy of Roles and Unified Workflows

The advent of LLMs has fundamentally reshaped the landscape of TSA, introducing new paradigms that extend far beyond the conventional modeling pipeline [34]. As summarized in Table I, we identify three principal roles of LLMs for TSA: **fine-tune-based inference engine (inference engine)**, **enhancer based on TSA methods (enhancer)**, and **hybrid collaborators**. for each category, we outline the underlying workflows and highlight key methodological advances, drawing on representative literature to illustrate the evolution and current directions in the integration of LLMs for TSA.

A. Fine-tune-based Inference Engine

In the role of inference engine, LLMs are directly adapted as the central model for TSA [24]. Compared with lightweight models, LLMs shows remarkable flexibility in learning temporal patterns, semantic information, and contextual dependencies via a unified architecture [23]. The workflows typically include four core components: tokenization approaches, prompt engineering, semantic alignment, and fine-tuning.

1) Tokenization Approaches: Tokenization for time series data should be regarded not merely as a preprocessing step but as a foundational representation learning mechanism that bridges continuous temporal signals with the discrete symbolic domain in which LLMs operate. Tokenization provides the essential foundation for LLMs to capture latent temporal structures and dependencies, thereby shaping their ability to model meaningful patterns and ultimately supporting downstream applications [1]. Within the paradigm of inference engines, tokenization serves as the entry point of workflows and directly conditions the effectiveness of subsequent components.

The mainstream tokenization approaches can be broadly categorized into three types: digit-level tokenization, patch-based tokenization, and bin-based tokenization. The general tokenization paradigm can be formulated as:

$$\mathbf{T} = \mathrm{Tokenize}(\mathbf{X};\, \theta) \tag{5}$$

where $\mathbf{X}$ denotes the original time series data, $\mathrm{Tokenize}(\cdot)$ is the tokenization function parameterized by strategy θ, and $\mathbf{T}$ represents the resulting sequence of tokens.

Digit-level Tokenization: Digit-level tokenization refers to treating each digit in a time series as an individual token, thereby enabling fine-grained representation of numerical values. This approach is particularly effective for discrete or categorical time series with limited value ranges. Representative examples include LLaMA2 [66], MedualTime [47], and PromptCast [37], all of which utilize digit-level tokenization to support prediction and dialogue tasks involving raw numerical data. However, its effectiveness diminishes considerably for long or high-resolution series, as it can significantly increase sequence length and reduce computational efficiency.

Patch-level Tokenization: Patch-level tokenization refers to segmenting the input time series into consecutive patches, with each patch regarded as a token. This paradigm was first applied to time series forecasting by PatchTST [16]. Building upon this foundation, models such as OFA [35], aLLM4TS [36], and TimeLLM [23] further aggregate contiguous time steps into patches, thereby enhancing the representation of temporal dependencies. UniTime [38] augments this strategy by randomly masking patches during training, which facilitates improved generalization. Patch-level tokenization has become a prevalent and effective technique, providing a balance between temporal granularity and sequence length. By treating subsequences as atomic units, this approach preserves local temporal structures and enables LLMs to efficiently process longer sequences. It is especially advantageous in scenarios with high-frequency or multi-scale data, where capturing dependencies over short intervals is essential.

Bin-level Tokenization: Chronos [55] introduces a bin-level scheme, transforming continuous and unbounded time series values into a finite vocabulary of tokens through scaling, quantization, and linguistic markers. This tokenization strategy not only constrains the input domain, but also enables LLMs to effectively capture and model the underlying temporal structures and dependencies present in time series data.

2) Prompt Engineering: Prompt engineering serves as a vital bridge that translates task requirements, domain knowledge, and contextual information into representations that LLMs can accurately interpret and leverage [18]. Within the TSA framework, prompt engineering functions as a key interface, converting domain-specific data, often processed through methods such as patch-level tokenization, together with auxiliary information, into formats compatible with LLM architectures [37]. This approach enhances the model's ability to capture latent temporal patterns and task-specific nuances inherent in TSA. Current research on prompt engineering for TSA primarily explores two paradigms: text-based prompts, which use natural language to specify tasks and data structures, and vector-based (soft) prompts, which encode domain knowledge as continuous vectors to provide LLMs with richer and more flexible contextual signals.

Text-based Prompt: Text-based prompts exhibit significant flexibility because they can be tailored through the design of various prompt templates. These prompts convert structured or numerical time series data into natural language or symbolic templates, often embedding statistical or contextual cues directly within the textual sequences. This approach enables the seamless integration of heterogeneous data such as domain knowledge, lagged features into a unified interface [37], [46], [56]. The general paradigm is defined as follows:

$$\mathbf{P} = \mathrm{TPrompt}(\mathbf{T}_X,\, \mathbf{T}_F) \tag{6}$$

where $\mathbf{T}_X$ denotes the tokenized representation of the original time series data $\mathbf{X}$, $\mathbf{T}_F$ denotes the tokenized representation of additional features or contextual information $\mathbf{F}$ (such as statistical indicators, lag features, or external context), and $\mathrm{TPrompt}(\cdot)$ is a text template function that organizes and combines these tokens to generate the final prompt $\mathbf{P}$.

TABLE I
CLASSIFICATION OF LLMS FOR TSA BY ROLE (I.E., INFERENCE ENGINE, ENHANCER, HYBRID COLLABORATOR) WITH DETAILED WORKFLOWS (I.E., TOKENIZATION, PROMPT, SEMANTIC ALIGNMENT, FINE-TUNING), TARGET TASKS, AND CODE AVAILABILITY.

Model	Role	Workflows				Task	Code
		Tokenization	Prompt	Semantic Alignment	Fine-tuning		
OFA [35]	Inference Engine	Patch-level	Vector-based	Emb-Injected	Endogenous (Direct)	General	Yes
aLLM4TS [36]	Inference Engine	Patch-level	Vector-based	-	Hybrid (Direct)	Forecasting	Yes
PromptCast [37]	Inference Engine	Digit-level	Text-based	-	-	Forecasting	Yes
TimeLLM [23]	Inference Engine	Patch-level	Vector-based	Emb-Injected	Exogenous (Direct)	Forecasting	Yes
UniTime [38]	Inference Engine	Patch-level	Vector-based	Emb-Injected	Exogenous (Direct)	Forecasting	Yes
AutoTimes [39]	Inference Engine	Patch-level	Vector-based	-	Hybrid (Direct)	Forecasting	Yes
S2IP-LLM [40]	Inference Engine	Patch-level	Vector-based	Distributional	Exogenous (Direct)	Forecasting	Yes
ETP [41]	Inference Engine	-	Vector-based	Contrastive	Exogenous (Direct)	Classification	No
TENT [42]	Inference Engine	-	Vector-based	Contrastive	Exogenous (Direct)	Classification	No
Qiu et al. [43]	Inference Engine	-	Vector-based	Distributional	-	Classification	No
MTAM [44]	Inference Engine	Patch-level	-	Distributional	-	Classification	No
TimeCMA [45]	Inference Engine	Digit-level	Text-based	Distributional	Exdogenous (Direct)	Forecasting	Yes
TableTime [46]	Inference Engine	Digit-level	Text-based	Distributional	Exdogenous (Direct)	Classification	Yes
MedualTime [47]	Inference Engine	Digit-level	Text-based	Emb-Injected	Exdogenous (Direct)	Classification	Yes
METS [48]	Enhancer	-	-	Contrastive	-	Classification	Yes
Xie et al. [49]	Enhancer	-	Text-based	-	Hybrid (Direct)	Forecasting	No
TimeReasoner [50]	Enhancer	-	Text-based	-	-	Forecasting	No
Time-R1 [51]	Enhancer	-	Text-based	-	-	Forecasting	No
Time-RA [52]	Enhancer	-	-	-	-	Anomaly Detection	Yes
TEMPO [53]	Inference Engine / Enhancer	Patch-level	Vector-based	Emb-Injected	Hybrid (LoRA)	Forecasting	Yes
LLM4TS [34]	Inference Engine / Enhancer	Patch-level	Vector-based	-	Endogenous (LoRA)	Forecasting	Yes
TEST [54]	Inference Engine / Enhancer	Patch-level	Vector-based	Contrastive	Exogenous (Direct)	General	No
Chronos [55]	Inference Engine / Enhancer	Bin-level	Vector-based	-	Exogenous (Direct)	General	Yes
LLM-Mob [56]	Inference Engine / Enhancer	Digit-level	Text-based	-	-	Forecasting	Yes
TimeCAP [57]	Inference Engine / Enhancer	-	Text-based	-	Exogenous (Direct)	Forecasting	Yes
TS-Reasoner [58]	Hybrid Collaborator	-	Text-based	-	-	Forecasting	No
AuxMobLCast [59]	Hybrid Collaborator	Digit-level	Vector-based	-	-	Forecasting	No
LAMP [60]	Hybrid Collaborator	-	Text-based	-	-	Forecasting	No
LA-GCN [61]	Hybrid Collaborator	Digit-level	Text-based	-	-	Forecasting	No
Chen et al. [62]	Hybrid Collaborator	-	Text-based	-	-	Forecasting	No
Park et al. [63]	Hybrid Collaborator	-	Text-based	-	-	Anomaly Detection	No
Zuo et al. [64]	Hybrid Collaborator	-	Text-based	-	-	Forecasting	No
DualSG [65]	Hybrid Collaborator	-	Text-based	-	Exogenous (Direct)	Forecasting	Yes

Recent studies have explored various directions for text-based prompts, highlighting their versatility. In time series forecasting and classification tasks, methods such as AuxMobLCast [59], LLM-Mob [56], and [67] utilize text-based prompts to incorporate domain knowledge, mobility patterns, and health-related features, respectively. Certain approaches, such as PromptCast [37], further enhance prompt informativeness by integrating lagged features and auxiliary signals. Additionally, LSTPrompt [68] introduces structured prompt designs, including task decomposition and adaptive reasoning strategies, to systematically guide LLMs in decision-making. Other works demonstrate the flexibility of text-based prompts in handling multimodal and heterogeneous information. For example, [69] combine textual news and price time series data for financial forecasting, while [49] show that historical price features can be extracted and embedded into prompts for zero-shot predictions. TableTime [46] reformulates time series classification as a table understanding problem by encoding time series data as tabular text, thereby improving information retention and facilitating integration with LLMs.

Vector-based (Soft) Prompt: Unlike text-based prompts, vector-based (soft) prompts operate directly in the embedding space, eliminating the need to convert time series data into textual representations [35], [47], [53]. These approaches can be broadly grouped into two categories:

Most studies focus on constructing learnable prompts that inject task-specific and temporal information into LLMs. In these approaches, prompt vectors are treated as trainable variables and integrated with time series embeddings. This design allows the model to better capture complex sequential dependencies and adapt to distributional shifts. The process can be formally expressed as:

$$\mathbf{P} = \mathrm{VPrompt}_{\theta}\big(\mathrm{Embed}(\mathbf{T}_X),\, \mathrm{Embed}(\mathbf{T}_F)\big) \tag{7}$$

where $\mathrm{VPrompt}_{\theta}$ denotes a learnable prompt module with parameters θ, which fuses the embedded time series tokens $\mathbf{T}_X$ and any additional contextual features $\mathbf{T}_F$.

Representative methods such as TEST [54], TEMPO [53], and LLM4TS [34] enhance model understanding and processing of time series data through innovative prompt design and encoding techniques. Recent approaches like S2IP-LLM [40], TimeLLM [23], and AutoTimes [39] further focus on semantic alignment by utilizing vector-based prompts to better align time series data with the input formats required by LLMs. These advanced methods not only effectively bridge modality gaps but also help ensure consistent representation of different data types, thereby optimizing performance in various time series tasks. The next section will explore these methods in

more detail, with a focus on their applications in semantic alignment and their critical role in multi-modal TSA.

3) Semantic Alignment: Semantic Alignment refers to the process of mapping time series representations into the semantic embedding space of LLMs, ensuring that temporal features become interpretable, manipulable, and predictive within the LLMs architecture [54]. The semantic alignment is essential to leverage LLMs' reasoning and generation capabilities for TSA, which allows downstream tasks to be performed using natural language prompts or decoders. Fundamentally, semantic alignment can be abstracted as a cross-modal alignment problem, where the objective is:

$$\min_{\phi,\theta} \mathcal{M}_{\text{align}}(g_\phi(\mathbf{X}), f_\theta(\mathbf{T})) \tag{8}$$

where g_ϕ is the mapping function that projects the time series data $\mathbf{X}$ into the semantic space as $\mathbf{Z}x = g_\phi(\mathbf{X})$, f_θ maps the text or language data $\mathbf{T}$ into the semantic space as $\mathbf{Z}t = f_\theta(\mathbf{T})$, and $\mathcal{M}_{\text{align}}(\mathbf{Z}_x, \mathbf{Z}_t)$ is the alignment objective ensuring that the representations satisfy the alignment constraints. Recent research on time series and text semantic alignment can be broadly categorized into three main approaches: **contrastive alignment**, **distributional alignment**, and **embedding-injected alignment**.

Contrastive Alignment: It leverages contrastive learning to jointly train time series and text encoders by maximizing the similarity of positive pairs (e.g., matched time series and textual descriptions) while minimizing that of negative pairs, as exemplified by the ETP [41]. Commonly, InfoNCE loss and similarity metrics such as inner product or cosine similarity are used. This approach has been widely applied in domains like healthcare and IoT, enabling the learning of structured semantic representations by pairing signals with corresponding texts. Further, models such as TEST [54] and TENT [42] employ multi-granularity alignment strategies to enhance semantic consistency and reasoning.

Distributional Alignment: It focuses on reducing the overall distributional discrepancy between time series data and text embeddings, rather than only optimizing pairwise similarity, as seen in methods like the OT-based framework for ECG-text alignment [43]. Representative techniques include optimal transport, canonical correlation analysis, and Wasserstein distance, all of which are particularly effective in scenarios with limited annotations or high noise. Applications include dual-branch encoding frameworks such as TimeCMA [45] and MedualTime [47], which jointly fuse raw time series data and text-derived features to enhance both robustness and generalization, as well as the use of distribution-based constraints in models like MTAM [44] and S2IP-LLM [40] to further improve cross-modal consistency in tasks such as medical report generation and emotion recognition.

Embedding-Injected Alignment: It introduces a time series encoder whose output is projected into a space compatible with LLMs, and then injected directly as input or context, as demonstrated by OFA [35]. This approach typically relies on structural alignment and prompt engineering, often freezing the LLM parameters and training only the projection module, without explicit contrastive or distributional losses. Time-LLM [23], UniTime [38], and TEMPO [53] employ strategies like segment embedding, prototype aggregation, and the incorporation of task descriptions and domain-specific prompts to enable multi-task and multi-granular modeling.

4) Fine-tuning: Fine-tuning of LLMs for TSA aims to adapt pre-trained models to specific tasks by optimizing a task-specific loss function [45]:

$$\min_{\mathbf{\Theta}} \mathcal{L}(\mathcal{F}(\mathbf{X}, \mathbf{\Theta}), \hat{\mathbf{Y}}, \boldsymbol{S}) \tag{9}$$

where $\mathbf{X}$ represents the input time series data, such as historical observations of financial or environmental variables; $\hat{\mathbf{Y}}$ is the corresponding target label, representing the expected output or prediction; $\mathbf{\Theta}$ denotes the optimized model parameters; $\mathcal{L}$ is the loss function quantifying the difference between the output of the LLMs $\mathcal{F}(\mathbf{X}, \mathbf{\Theta})$ and the true target labels $\hat{\mathbf{Y}}$, influenced by the fine-tuning strategy $\boldsymbol{S}$.

Fine-tuning of LLMs for TSA can be analyzed from two perspectives: parameter selection (i.e., select the parameters ($\mathbf{\Theta}$) for updating), and fine-tuning strategie selection (select the fine-tuning strategies ($\mathcal{S}$) for updating). These two aspects work together to optimize LLMs.

Parameters ($\mathbf{\Theta}$) Selection: Fine-tuning strategies can be categorized into three types according to which parameters are updated during fine-tuning: **endogenous parameters** ($\mathbf{\Theta}_{\text{en}}$), **exogenous parameters** ($\mathbf{\Theta}_{\text{ex}}$), and **hybrid parameters** ($\mathbf{\Theta}_{\text{h}}$).

$\mathbf{\Theta}_{\text{en}}$ refers to the internal parameters of pre-trained LLMs, including attention weights ($\mathbf{\Theta}_{\text{attn}}$), FFN weights ($\mathbf{\Theta}_{\text{ffn}}$), and layer normalization parameters ($\mathbf{\Theta}_{\text{ln}}$), which together form the core representational and reasoning capabilities of the model. Methods such as OFA [35], MedualTime [47] and LLM4TS [34] adopt this approach by selectively fine-tuning specific internal components to enable the model to better adapt to time series tasks. However, since the pre-trained LLM already contains a wealth of general knowledge, inappropriate modification of these internal parameters may disrupt existing representations and reduce generalization, especially when facing complex or low-resource scenarios.

$\mathbf{\Theta}_{\text{ex}}$ refers to parameters other than the internal parameters of the LLM that are kept fixed, including prompt parameters ($\mathbf{\Theta}_{\text{prompt}}$) and expanded layers ($\mathbf{\Theta}_{\text{expansion}}$). Adjusting these parameters does not disrupt the inherent knowledge of the LLM itself. S2IP-LLM [40] and Time-LLM [23] freezes the layers of the pre-trained LLM and introduces two new learnable modules: Patch Reprogramming and Output Projection, applying direct fine-tuning to these new modules. TEST [54] fine-tunes vector-based soft prompts to enable the model to better capture temporal patterns in time series data. However, it can be challenging due to the complexity of aligning these additional components with task-specific requirements. The main difficulty lies in ensuring that the adapted parameters interact harmoniously with the pre-trained representations, as improper alignment may lead to suboptimal performance.

Some recent approaches such as aLLM4TS [36], TEMPO [53] and AutoTimes [39] update $\mathbf{\Theta}_{\text{h}}$ (i.e., hybrid parameters including endogenous and exogenous parameters), aiming to leverage the advantages of both strategies. This method enables flexible adaptation and often achieves better task alignment, but requires careful coordination to avoid conflicting updates and maintain model stability.

Fine-tuning Strategies ($\mathcal{S}$) Selection: Fine-tuning strategies can also be distinguished by the method used to update the model parameters. The two primary methods are **direct fine-tuning ($\mathcal{S}_{\mathbf{d}}$)** and **LoRA-based fine-tuning ($\mathcal{S}_{\mathbf{lora}}$)**.

$\mathcal{S}_{\mathrm{d}}$ refers to the process where specific layers of the pre-trained LLM are updated during the fine-tuning process, such as Time-LLM [23] and TEST [54]. A potential drawback is that this method can be less efficient, as the layers that are typically fine-tuned often have a large number of parameters, which increases the computational cost.

$\mathcal{S}_{\mathrm{lora}}$ introduces low-rank matrices to specific layers of the LLM, leaving the core parameters of the model frozen. Only the low-rank matrices are updated, enabling task-specific adaptation without modifying the original pre-trained weights. Representative works such as aLLM4TS [36], LLM4TS [34] and TEMPO [53] employ this parameter-efficient fine-tuning strategy. This method facilitates efficient fine-tuning while preserving the learned representations of the pre-trained model.

B. Enhancer based on TSA Methods

In contrast to fine-tune-based inference engines, where LLMs serve as primary predictive modules, the enhancer role positions LLMs as auxiliary components in TSA workflows. In this framework, LLMs support traditional TSA methods through strategies such as enhancing time series data with self-supervised learning, synthetic data generation, and multi-modal processing, as well as actively promoting model interpretability using advanced reasoning techniques like chain-of-thought (CoT) prompting and explanation generation.

1) Enhancement of Time Series Data: The quality and informativeness of input data are critical factors affecting the performance of TSA. Recent studies show that LLMs, when combined with advanced data enhancement strategies, can significantly improve the robustness and predictive accuracy of time series models [55]. Three principal approaches have proven particularly effective: self-supervised learning, synthetic data generation, and multi-modal data fusion.

Self-Supervised Learning. Self-supervised learning techniques enable LLMs to collaborate with methods that enhance time series data by learning patterns and dependencies without requiring labeled data. For example, models like LLM4TS [34] introduce the two-stage fine-tuning strategy. In the first stage, self-supervised learning methods are applied to align patch-based tokens with the semantic space of LLMs. This approach allows the model to uncover hidden temporal patterns and relationships in the data, thereby improving performance on forecasting task. aLLM4TS [36] redefines time series forecasting as a self-supervised multi-patch prediction task. This conceptual shift allows the model to predict multiple segments of time series simultaneously, further enhancing its ability to understand and forecast complex temporal dependencies.

Synthetic Data Generation. Synthetic data generation has become an essential tool for improving the generalization and robustness of TSA. By artificially expanding the diversity and coverage of training datasets, these methods help mitigate challenges such as data scarcity, imbalance, and overfitting. For example, Chronos [55] demonstrates the advantages of combining mixup with kernel-based synthesis. Meanwhile, LLMs can serve as contextualization agents, generating enriched summaries or intermediate representations to further enhance the informativeness of synthetic data [57]. These developments collectively reinforce the importance of synthetic data generation in supporting robust and interpretable TSA.

Multi-Modal Data Fusion. Multi-modal data fusion has become an important strategy for advancing TSA by integrating structured numerical data with unstructured textual or contextual information [70]. This method leverages the strengths of different data modalities, enabling models to capture more comprehensive patterns and richer contextual signals. For example, [49] shows that providing ChatGPT with both historical stock prices and related tweets can improve zero-shot stock movement prediction, illustrating the effectiveness of textual data integration. METS [48] further demonstrates that combining ClinicalBert with both ECG signals and medical report texts enhances clinical event classification by expanding the semantic space of the model. TEMPO [53] employs generative transformers to extract semantic embeddings from news reports and incorporates these embeddings as learnable prompts together with time series inputs. These representative studies clearly highlight the increasing importance of multi-modal data fusion for providing deeper context, improving interpretability, and supporting more robust and generalizable TSA in complex, dynamic real-world environments.

2) Enhancement of Interpretability: Interpretability remains a significant challenge in deep learning-based TSA. The development of CoT reasoning [71] has introduced innovative new methods for enhancing model transparency and explainability. CoT is a prompting technique that encourages LLMs to generate intermediate reasoning steps during predictions. This process facilitates a deeper and more structured understanding of model behavior, making predictions considerably more interpretable and transparent. Several recent models have demonstrated the practical benefits of integrating interpretability mechanisms into time series tasks. For example, TimeReasoner [50] leverages slow-thinking LLMs with CoT prompting to produce highly interpretable and reliable forecasts, underscoring the potential of LLMs as transparent and accountable analytical tools. Time-R1 [51] implements a two-stage framework in which the model first learns multi-step temporal reasoning through supervised CoT fine-tuning, followed by reinforcement learning based on both interpretability and accuracy. LLM-Mob [56] uses ChatGPT to detect anomalous human mobility patterns and provide additional insights into model decision processes. In the context of time series anomaly analysis, Time-RA [52] presents a comprehensive benchmark integrating detection, localization, and causal reasoning across multi-modal data, employing structured prompt-based annotation and AI-assisted feedback pipelines to ensure high-quality, interpretable, and robust outputs.

C. Hybrid Collaborator

In addition to the aforementioned roles of LLMs as fine-tune-based inference engines and data enhancers, we classify all other approaches that leverage LLMs in combination

with external modules as hybrid collaborators. This category encompasses a broad range of emerging methods that integrate frozen or partially fine-tuned LLMs with traditional lightweight models. Such hybrid collaborators typically facilitate the synergistic use of LLMs for feature augmentation, intermediate reasoning, or decision support, thereby enabling more versatile and adaptive TSA pipelines. These frameworks not only improve predictive accuracy and computational efficiency, but also extend the applicability of LLMs to complex scenarios where multiple models or expert modules need to be coordinated within the analysis process.

Representative studies illustrate the diversity of hybrid collaboration strategies. For example, TS-Reasoner [58] employs LLM-generated programs to decompose and orchestrate multi-step reasoning tasks, integrating outputs from traditional time series models and numerical methods to solve complex forecasting objectives. Multi-stage frameworks such as Aux-MobLCast [59] and related approaches such as LAMP [60] incorporate frozen LLMs as intermediate processors, enabling the progressive refinement of features and predictions through the sequential combination of LLMs and neural network modules. In the context of multivariate time series, models like LA-GCN [61] and [62] utilize LLMs to uncover inter-series dependencies or extract structural information from domain texts, subsequently integrating these insights with graph neural networks for improved predictive performance. DualSG [65] introduces a semantic-guided dual-stream framework for multivariate time series forecasting, where an LLM serves as a semantic guidance module. The LLM generates natural language time series captions to enrich numerical predictions with interpretable context, and a description-guided fusion module explicitly models variable relationships, enhancing both accuracy and interpretability. Hybrid frameworks are also being explored in financial applications. Park [63] demonstrates the benefits of combining LLMs with statistical models and multi-agent systems for automated and precise anomaly detection in financial markets. LA-GCN [64] propose an agent-based collaborative architecture where multiple specialized agents, coordinated by LLMs, systematically bridge human expertise and algorithmic forecasting. This design lowers the barrier for non-expert users and enables the seamless incorporation of domain knowledge throughout the analytical pipeline.

IV. Application

In this section, we systematically review how LLMs empower TSA across various domains, as illustrated in Figure 2. By effectively leveraging their advanced capabilities in processing complex and heterogeneous data, LLMs have driven significant progress in a wide range of industries such as finance, transportation, energy, and others.

A. Finance

The domain of financial TSA is characterized by profound complexity, stemming from its reliance on diverse and heterogeneous data sources. These inputs range from structured numerical data, such as market prices and trading volumes, to unstructured information derived from news articles, corporate filings, and social media sentiment [72]. This data is typically noisy, non-stationary, and highly interdependent, exhibiting periodic fluctuations, short-term volatility, and long-term trends, all while being susceptible to exogenous shocks from policy changes and global events [73]. The advent of LLMs presents a paradigm shift, offering novel solutions to these long-standing challenges and poised to revolutionize financial analysis [74], [75].

1) Stock Movement (Trend) Forecasting: Stock movement (trend) forecasting refers to the prediction of future price directions or trends in financial markets based on historical and real-time data. Accurate forecasting is essential for quantitative trading, risk management, and market sentiment analysis, as it forms the foundation for informed investment decisions and effective financial strategies. LLMs enhance this capability through several key mechanisms:

Multi-modal Data Fusion: LLMs facilitate the integration of multimodal financial data, combining unstructured textual sources such as news articles and social media with structured time series data. Representative frameworks, including [62], LG+SCRL [76], and Ploutos [77], utilize LLMs to extract dynamic inter-company relationships from textual information and merge these insights with quantitative features. This approach enhances the model's ability to capture market events, global trends, and complex interactions among companies.

Zero-shot Learning: LLMs demonstrate promising zero-shot and few-shot capabilities, allowing for trend prediction without domain-specific fine-tuning. Works like [49] benchmark ChatGPT's ability to generalize across modalities (e.g., tweets and price data), revealing both its potential and current limitations in unsupervised financial forecasting.

Decision Support: Fine-tuned LLMs and hybrid frameworks leverage rich text representations to optimize investment decisions and portfolio construction. Studies such as Fine-Tuning LLMs for Stock Return Prediction [78], Ploutos [77], and LG+SCRL [76] show that LLM-driven features and reasoning can outperform traditional sentiment scores, boost returns, and reduce risk in real-world financial markets.

Interpretability Enhancement: LLMs empowered by prompt engineering can automatically extract interpretable factors and causal logic from financial texts, thereby generating natural language explanations for stock movement predictions. Approaches such as LLMFactor [79], SEP [80], and Ploutos [77] leverage knowledge-guided prompting and autonomous rationale generation to identify key drivers of stock movements and provide human-readable, context-aware justifications. This enhances the interpretability, transparency, and trustworthiness of model predictions, supporting both informed decision-making and regulatory compliance.

2) Stock Price Forecasting: Stock price forecasting aims to predict the prospective returns of financial assets, a central component of investment decision-making and portfolio optimization. LLMs contribute by:

Multi-modal Data Fusion: LLMs can deeply integrate financial time series data with text, company metadata, news, images, and other modalities, enhancing the model's understanding of market dynamics and interpretability. Representa-

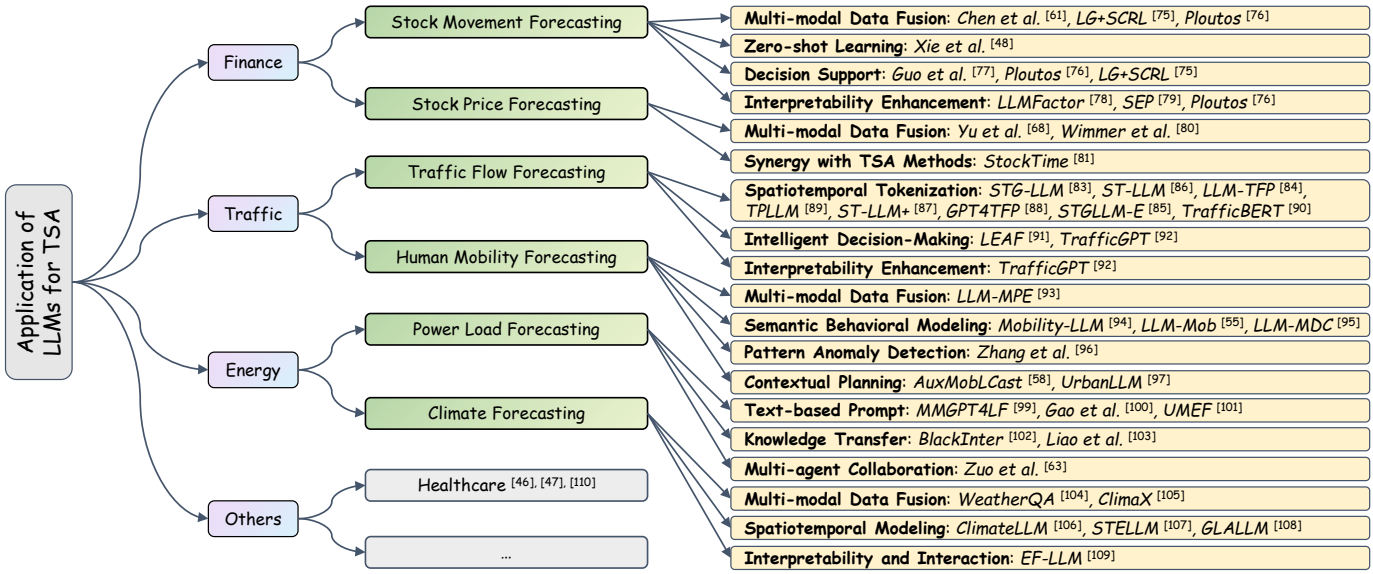

Fig. 2. An overview of downstream domains empowered by LLMs in TSA, including finance, traffic, energy, and other fields, together with their corresponding tasks. Specifically, we highlight key techniques enabled by LLMs, such as multimodal data fusion, interpretability enhancement, and multi-agent collaboration. By leveraging these techniques, LLMs significantly advance the breadth and depth of applications and tasks across various TSA domains.

tive frameworks such as [69] unify historical prices, company information, and news text as input for LLMs, enabling explainable predictions for NASDAQ-100 stock prices. Leveraging vision-language models [81] encodes price data as images and bytes, using vision-language models to capture fine-grained market fluctuations and achieve superior prediction accuracy compared to traditional approaches.

Synergy with TSA Methods: Integrating LLMs with established TSA techniques allows for a more robust capture of financial data's structural properties like periodicity and trends. Approaches such as StockTime [82] and Time Series Forecasting with LLMs leverage LLMs to process price sequences as tokens and incorporate statistical features, thereby improving prediction accuracy and model robustness for financial TSA.

B. Traffic

Traffic datasets serve as a representative example of spatiotemporal data, as they incorporate both temporal and spatial dependencies. Temporal patterns may involve daily cycles, abrupt changes due to incidents or weather conditions, and long-term trends. Spatial dependencies arise from the connectivity of road networks and the influence of neighboring locations. These complex patterns are typically captured by sensors such as cameras and GPS devices [83].

1) Traffic Flow Forecasting: Accurate traffic flow forecasting is crucial for urban and traffic management. LLMs enhance forecasting through several innovative mechanisms:

Spatiotemporal Tokenization: The introduction of spatiotemporal tokenizers and encoders allows for the conversion of diverse traffic data, such as road network topology, sensor measurements, and temporal sequences, into unified token representations compatible with LLM architectures. This methodology enables effective modeling of the intrinsic spatiotemporal structures and dependencies present in traffic datasets, thereby enhancing the ability of LLMs to capture complex patterns and interactions within transportation systems. Representative spatiotemporal modeling frameworks employ various strategies to adapt LLMs for traffic flow prediction. For instance, STG-LLM [84], LLM-TFP [85], and STGLLM-E [86] utilize customized tokenization schemes and adapter modules to encode graph-based spatial relationships and temporal dynamics, enabling LLMs to capture complex interactions across large-scale road networks. ST-LLM [87] and ST-LLM+ [88] leverage spatiotemporal embeddings and prompt-based mechanisms to encode traffic points and sequences as tokens, enhancing the LLM's ability to model fine-grained spatial-temporal dependencies. Models such as GPT4TFP [89] and TPLLM [90] integrate multi-head attention and cross-modality fusion techniques, facilitating deep integration of spatial and temporal features. Additionally, TrafficBERT [91] adopts a pretraining paradigm based on BERT architecture, employing multi-head self-attention to learn robust representations for traffic flow prediction tasks.

Intelligent Decision-Making: LLMs serve as sophisticated high-level decision modules capable of selecting, integrating, and ranking outputs from multiple model branches or prediction sources. This capability enhances the robustness and adaptability of traffic forecasting systems under diverse and dynamic traffic conditions. For instance, LEAF [92] utilizes LLMs to select the optimal prediction from graph and hypergraph branches during inference, while TrafficGPT [93] orchestrates multiple traffic base models to decompose complex tasks and provide strategic recommendations.

Interpretability Enhancement: LLMs enhance interpretability in traffic forecasting systems by enabling natural language interaction, generating explanations, and supporting decision-making. For example, TrafficGPT [93] allows users to engage in dialogue, decompose tasks, and iteratively refine

results based on feedback.

2) Human Mobility Forecasting: Human mobility prediction serves as a cornerstone for urban planning, event management, and intelligent transportation systems, requiring robust modeling of complex behavioral patterns across heterogeneous data sources [2]. LLMs are driving transformative progress by:

Multi-modal Data Fusion: LLMs employ advanced prompt engineering to transform diverse mobility-related information, such as raw trajectory data, event descriptions, POI attributes, and social media feeds, into unified natural language representations. This approach enables LLMs to fuse structured and unstructured data, capturing nuanced behavioral and contextual dependencies. For example, frameworks like LLM-MPE [94] leverage LLMs to process unstructured public event descriptions and integrate them with historical mobility data.

Semantic Behavioral Modeling: By reprogramming mobility sequences as natural language prompts, LLMs are empowered to extract and model latent behavioral intentions and travel preferences. Approaches like Mobility-LLM [95] and LLM-Mob [56] employ memory networks and prompt pools to guide LLMs in interpreting check-in sequences, thereby uncovering users' visiting intentions and travel patterns. LLM-MDC [96], which proposes an interactive and step-by-step next location prediction framework driven by LLMs. By leveraging multi-round dialogue, activity-aware prompts, and candidate set enhancement, the model can semantically reason about user mobility behaviors and provide explainable predictions. This capability advances personalized mobility prediction and enhances the interpretability of model outputs.

Pattern Anomaly Detection: LLMs demonstrate strong generalization and reasoning abilities in detecting anomalous mobility behaviors and mining spatiotemporal patterns. Through prompt engineering and contextual querying, LLM-based frameworks such as [97] show that LLMs can identify irregularities in human trajectories and provide human-readable explanations, improving transparency and reliability in applications like public safety and health monitoring.

Contextual Planning: LLMs enable context-aware mobility forecasting by integrating temporal, spatial, and semantic information within a unified modeling framework. Systems such as AuxMobLCast [59] and UrbanLLM [98] leverage LLMs to decompose complex urban tasks, coordinate with specialized AI models, and generate comprehensive solutions for activity planning and management.

C. Energy

Energy data exhibits significant heterogeneity and complex spatiotemporal dependencies. Traditional statistical and machine learning approaches often encounter challenges in fusing data from multiple sources, adapting to distributional shifts, and modeling spatiotemporal correlations [99].

1) Power Load Forecasting: Power load forecasting is crucial for stable operation and planning of power systems, requiring accurate predictions of electricity demand across temporal and spatial scales. LLMs contribute through:

Text-based Prompt: Text-based prompt engineering facilitates the transformation of structured spatiotemporal data, such as historical load sequences, meteorological observations, equipment status, and textual reports, into unified natural language representations. This approach enables LLMs to process and fuse heterogeneous modalities within a shared semantic space. Representative models such as MMGPT4LF [100], [101], and UMEF [102] demonstrate that this paradigm enhances the modeling of complex dependencies and improves the robustness of predictions, particularly in scenarios with incomplete or noisy data.

Knowledge Transfer: Pretrained LLMs exhibit substantial cross-domain knowledge, facilitating swift adaptation in scenarios with limited or novel data. For example, BlackInter [103] employs inductive plug-in adapters, enabling scalable deployment across various building types without the necessity of accessing the underlying model architecture or parameters. This approach leverages the inherent generalization capability of LLMs and does not require prior domain-specific pre-training or fine-tuning on the target data. Furthermore, [104] empowered as a generative pre-trained transformer for time series, exhibits strong generalization performance through transfer learning, even when historical load data is scarce.

Multi-agent Collaboration: LLMs can also function as intelligent agents in multi-agent frameworks for power load forecasting [64]. By enabling system operators to use natural language instructions, such frameworks integrate expert knowledge and experience into forecasting. This not only streamlines workflows for non-experts but also improves prediction accuracy and supports real-world deployment.

2) Climate (Weather) Forecasting: Climate and weather forecasting aims to predict atmospheric and environmental conditions over short and long time horizons, supporting disaster prevention, resource management, and societal planning. Their contributions include:

Multi-modal Data Fusion: LLMs unify numerical meteorological observations, remote sensing imagery, and textual forecasts within a single modeling framework, thereby enhancing their ability to reason about extreme weather and climate events. Benchmarks such as WeatherQA [105] and unified architectures like ClimaX [106] have demonstrated the effectiveness of multimodal integration in supporting complex climate reasoning tasks.

Spatiotemporal Modeling: By combining techniques such as frequency domain decomposition, mixture-of-experts mechanisms, and global-local feature fusion, LLMs can make more accurate predictions of complex climate systems and extreme weather events. For instance, ClimateLLM [107] combines Fourier frequency domain decomposition with a Mixture-of-Experts module to efficiently model multi-frequency meteorological signals; STELLM [108] employs seasonal-trend decomposition and spatiotemporal prompt mechanisms to enhance LLMs' representation of complex wind speed patterns; and GLALLM [109] integrates the global sequence modeling strengths of LLMs with the local spatial correlation learning of hypergraph neural networks, achieving effective information fusion through a meta-network.

Interpretability and Interaction: LLMs improve energy and weather forecasting by offering natural language explanations, generating risk alerts, and supporting interactive

decision-making. These features make forecasting systems easier to use and build trust with non-experts. For example, EF-LLM [110] streamlines the entire forecasting process through human-AI interaction, providing interpretable results and practical insights. This reduces dependence on expert engineers and enables more efficient, user-friendly forecasting.

D. Others

LLMs are driving significant advances in healthcare time series analysis by leveraging their powerful capabilities in processing multimodal data and generating transparent, natural language explanations. These strengths are especially valuable in clinical contexts, where both interpretability and the integration of heterogeneous information are essential for effective decision-making. For example, the METS framework [48] combines deep learning-based ECG encoders with clinical language models, enabling self-supervised classification that aligns physiological signals with textual clinical records. Similarly, NYUTron [111] utilizes large-scale electronic health record data to train clinical LLMs, thereby achieving strong performance across a range of prediction tasks. Collectively, these approaches highlight the unique ability of LLMs to bridge numerical and textual data, deliver patient-specific interpretations, and generalize effectively across diverse clinical scenarios. This progress addresses key limitations of traditional black-box clinical models and lays the foundation for more interpretable and reliable healthcare analytics. Recent work, such as MedualTime [47], further demonstrates the expanding impact of LLMs in this field.

V. Existing Challenges and Future Directions

The application of LLMs for TSA has created new opportunities across various domains. This section systematically reviews the major obstacles, such as cross-domain generalization, privacy preservation, and multimodal data alignment, and outlines promising future directions for addressing these issues [112]. By critically examining both the limitations of current methodologies and the opportunities for methodological innovation, we aim to provide a comprehensive roadmap for future research and practical deployment of LLMs in TSA.

A. Cross-Domain Generalization

1) Existing Challenges: Time series data from different domains often exhibit significant distributional disparities and pronounced heterogeneity in the number and semantics of variables and features. Mainstream approaches predominantly rely on fine-tuning pre-trained LLMs separately on each dataset or domain. To address these challenges, recent studies have introduced strategies such as channel-wise independence [16] and reversible normalization (RevIN) [113], which partially mitigate the effects of distribution shifts and variable inconsistency during training. However, these methods remain fundamentally limited, as their focus on isolated domain adaptation fails to fully explore the potential of unified models with intrinsic generalization capabilities across diverse domains and feature spaces. Recent advancements, including extensions of frameworks such as UniTime [38] and LangTime [114], have initiated efforts to align heterogeneous time series representations, design more expressive prompting mechanisms, and integrate adaptive learning techniques. These approaches aim to accommodate varying data distributions, temporal resolutions, and feature sets across domains.

2) Future Directions: Future research should focus on developing FMs for time series that incorporate large-scale multi-domain pre-training, continual learning, and advanced meta-learning paradigms. These strategies are essential for equipping LLMs with the ability to generalize flexibly and reliably across highly diverse data distributions, variable types, and contextual scenarios encountered in real-world applications. Achieving universal adaptability is critical for fully realizing the full potential of LLMs as foundational models for TSA and for enabling the development of practical and scalable solutions that address a wide range of temporal modeling tasks in heterogeneous and dynamic environments.

B. Privacy

1) Existing Challenges: Time series data often contain sensitive information, such as user behaviors, medical records, or financial transactions. The increasing adoption of LLM-based frameworks in TSA amplifies privacy concerns, as these models are frequently trained on large, potentially confidential datasets [25]. Most existing LLMs are designed with the assumption of centralized data access, which is often infeasible due to privacy regulations such as GDPR and HIPAA, as well as organizational policies. Additional risks, including model inversion, membership inference attacks, and unintended memorization of sensitive data, further threaten privacy and security. Common data anonymization and aggregation techniques may compromise modeling performance, resulting in suboptimal outcomes in critical scenarios [115].

2) Future Directions: Addressing privacy challenges in LLM-based TSA requires integration of privacy-preserving techniques. Key research directions include federated learning, which enables decentralized training without sharing raw data, and differential privacy, which provides formal protection at the model update level. Techniques such as secure multi-party computation and edge deployment further reduce data exposure risks. In addition, synthetic data generation methods based on the SEAL methodology support collaborative modeling while maintaining privacy. Progress in these areas is essential for developing trustworthy, regulation-compliant LLM solutions for sensitive time series applications.

C. Multimodal Data Alignment

1) Existing Challenges: LLMs are primarily pre-trained on large-scale text corpora, which grants them strong capabilities in natural language understanding and generation. However, this textual bias limits their compatibility with structured data such as time series or multimodal inputs [23]. To address this limitation, current approaches often employ text-based prompts or semantic alignment strategies (as shown in Section III) to convert time series and multimodal features into forms processable by LLMs. Despite recent progress, these

methods frequently struggle to capture complex cross-modal dependencies and may result in superficial semantic correspondence, causing potential information loss. Furthermore, the generalizability of such alignment strategies across domains and tasks remains uncertain. Inadequate alignment can also undermine the interpretability and robustness of LLM-based models, especially when faced with noisy or incomplete data.

2) Future Directions: Advancing multimodal TSA with LLMs requires the development of more adaptive and theoretically robust alignment methods. Future research should prioritize the design of advanced cross-modal encoders or alignment modules that explicitly model semantic relationships between modalities and language, moving beyond basic pseudo-text transformation or shallow fusion approaches. Techniques such as contrastive learning, co-attention mechanisms, and graph-based modeling offer promising avenues for achieving deeper and more effective alignment. Additionally, leveraging FMs trained on multimodal corpora and implementing context-aware prompting or dynamic fusion strategies can further enhance model scalability and generalization.

VI. Conclusion

In this survey, we systematically categorize the roles of LLMs for TSA, establishing a taxonomy of roles that includes fine-tune-based inference engine, enhancer based on TSA methods, and hybrid collaborators. Building on this comprehensive framework, we critically analyze how LLMs empower downstream domains such as finance, traffic, and energy. Specifically, we summarize the mechanisms by which LLMs enable domain-specific tasks, including multi-modal data fusion, decision support, and interpretability improvement. These mechanisms collectively illustrate the significant empowerment LLMs provide for advancing TSA methodologies and practical applications. Despite these advancements, challenges remain in critical areas such as cross-domain generalization, privacy protection, and multi-modal data alignment. Addressing these issues is essential for unlocking the full potential of LLMs in TSA. We hope the insights and perspectives presented in this survey will support further research and practical innovation in the field.

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